

$$(a) \bar{X} = \frac{X_1 + X_2 + X_3}{3}$$

$$E(\bar{X}) = E\left(\frac{X_1 + X_2 + X_3}{3}\right)$$

$$= \frac{1}{3} E(X_1 + X_2 + X_3) = \frac{1}{3} [E(X_1) + E(X_2) + E(X_3)] = \frac{1}{3} (\mu + \mu + \mu) = \mu$$

$$Var(\bar{X}) = Var\left(\frac{X_1 + X_2 + X_3}{3}\right) = \frac{1}{9} Var(X_1 + X_2 + X_3) = \frac{1}{9} [Var(X_1) + Var(X_2) + Var(X_3)]$$

$$= \frac{1}{9} (\sigma^2 + \sigma^2 + \sigma^2) = \frac{3\sigma^2}{9} = 0.333\sigma^2$$

$$(b) W = \frac{1}{6}X_1 + \frac{1}{6}X_2 + \frac{2}{3}X_3$$

$$E(W) = E\left(\frac{1}{6}X_1 + \frac{1}{6}X_2 + \frac{2}{3}X_3\right) = \frac{1}{6}E(X_1) + \frac{1}{6}E(X_2) + \frac{2}{3}E(X_3) = \frac{1}{6}\mu + \frac{1}{6}\mu + \frac{2}{3}\mu = \mu$$

$$Var(W) = Var\left(\frac{1}{6}X_1 + \frac{1}{6}X_2 + \frac{2}{3}X_3\right) = \frac{\sigma^2}{36} + \frac{\sigma^2}{36} + \frac{4\sigma^2}{9} = 0.5\sigma^2$$

(c) Clearly, $\bar{X}$ is better as it is unbiased, and its variance is smaller. For the important sampling distribution graphs, please refer to the tut discussion or supplementary notes.

(d) Discussed during the tuts.

2.

(a) Our objective is clearly to minimize the sum of squared residuals:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n \hat{u}_i^2 = \min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0} = 2 \times \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \times (-1)$$

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1} = 2 \times \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \times (-x_i)$$

$$\text{Let } \frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0} = 0 \text{---(1) and } \frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1} = 0 \text{---(2).}$$

Solve (1) and (2),

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \text{-----(1)}$$

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \times (x_i) = 0 \text{---(2)}$$

We can get the estimators $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Hessian matrix (for second partial derivative test) is

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_0} = 2 \sum_{i=1}^n (1), \quad \frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_1} = 2 \sum_{i=1}^n (x_i)$$

$$\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1 \partial \hat{\beta}_0} = 2 \sum_{i=1}^n (x_i), \quad \frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1 \partial \hat{\beta}_1} = 2 \sum_{i=1}^n (x_i^2)$$

Because $\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_0 \partial \hat{\beta}_0} > 0$, $\frac{\partial \sum_{i=1}^n \hat{u}_i^2}{\partial \hat{\beta}_1 \partial \hat{\beta}_1} = 2 \sum_{i=1}^n (x_i^2) > 0$ and

$$2 \sum_{i=1}^n (1) \times 2 \sum_{i=1}^n (x_i^2) - 2 \sum_{i=1}^n (x_i) \times 2 \sum_{i=1}^n (x_i) > 0,$$

The estimators indeed minimize the objective function.

(b)

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Plug $y_i = \beta_0 + \beta_1 x_i + u_i$ into the numerator,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})\beta_0}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})\beta_1 x_i}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \frac{\beta_0 \sum_{i=1}^n (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} + \beta_1 \frac{\sum_{i=1}^n (x_i - \bar{x})x_i}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= 0 + \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \beta_1 + \frac{\sum_{i=1}^n (x_i - \bar{x})u_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Take conditional expectation on both sides, the expectation of $\hat{\beta}_1$,

$$E[\hat{\beta}_1|x] = \beta_1 + E\left[\frac{\sum_{i=1}^n(x_i - \bar{x})u_i}{\sum_{i=1}^n(x_i - \bar{x})^2} | x\right] = \beta_1 + \frac{\sum_{i=1}^n(x_i - \bar{x})E[u_i|x]}{\sum_{i=1}^n(x_i - \bar{x})^2}$$

Under the zero conditional mean assumption $E[u_i|x] = 0$,

$$E[\hat{\beta}_1|x] = \beta_1, \text{ i.e., } E[\hat{\beta}_1] = \beta_1.$$

Note: Use the law of iterated expectation; take expectation on both sides.

$$E\{E[\hat{\beta}_1|x]\} = E(\beta_1)$$

(c) We don't expect $\hat{\beta}_1 = \beta_1$. However, $E(\hat{\beta}_1) = \beta_1$

(d) It can help us quantify, on average, how far away the $\hat{\beta}_1$ estimate is from the true value. We can use this formation to assess the reliability of $\hat{\beta}$, construct confidence interval and performance hypothesis testing.